DRO — Disaster Response Optimizer

SEA Quantathon 2026 | Vietnam | AI + Quantum Decision Support

Pitch: A web dashboard combining AI risk scoring with quantum and quantum-inspired optimization to help Vietnam and Southeast Asia response teams prioritize victims and assign rescue units faster in simulation and planning.

Scenario Data → **AI Risk Scoring** → **QUBO Optimizer** → **Dispatch Dashboard**

Problem

Floods, typhoons, landslides, and urban emergencies strain limited rescue teams, hospitals, and road networks across Central Vietnam and the wider region. Coordinators receive scattered SOS reports and hazard updates but lack one tool to rank high-risk cases and solve multi-team assignment under hard constraints—especially when connectivity is poor and decisions must be made in minutes.

Proposed Solution

DRO ingests simulated or live disaster data (victims, teams, road risk, hospital capacity, flood severity) and produces a prioritized response plan: victim/area rankings, team assignments, suggested routes, and estimated response times on an interactive map-based dashboard for dispatchers and field leads.

How AI Is Used

- **Risk scoring:** Gradient-boosted or rule-augmented models rank victims/areas (injury, isolation, flood depth, time since contact).
- **SOS text classification:** NLP parses unstructured reports (Vietnamese/English) into structured optimizer fields.

How Quantum Computing Is Used

Disaster dispatch is a **QUBO**: binary team–victim assignment variables; objective minimizes weighted response time and penalizes unmet high-risk demand; constraints encode capacity and unsafe edges. Solved via **QAOA**, **quantum annealing** (D-Wave/simulators), or **quantum-inspired simulated annealing** for demos without hardware.

Product Demo / Web App Flow

Central Vietnam scenario: Thua Thien Hue flood and landslide response simulation.

1. Select scenario; load victims, teams, hospitals, road risk.
2. Run AI risk scoring → heatmap updates.
3. Run optimizer → priority list, assignments, routes.
4. Compare baseline vs. optimized estimated response time.
5. Export plan summary for briefing.

Expected Impact

Supports faster coordinator decision-making; helps prioritize high-risk cases; can reduce *estimated* response time in simulation compared with a greedy baseline; improves transparency for multi-agency drills aligned with **SDG 3** (Health) and **SDG 11** (Sustainable Cities). *Simulation and drill support only—not a substitute for human command authority.*

Closing

DRO bridges AI situational awareness and quantum optimization for the combinatorial heart of disaster dispatch—built for Southeast Asia's realities, honest about simulation limits, and ready to grow with partners who save planning time when every minute counts.

2-Year Roadmap (M1–M24)

M1	Core data model; Thua Thien Hue scenario pack; simulated victims/teams/roads	**M2**	AI risk scoring MVP; priority queue and heatmap output	**M3**	QUBO formulation; quantum-inspired annealing; greedy baseline benchmark
M4	Streamlit dashboard shell; Pydeck map layer	**M5**	Thua Thien Hue scenario wired to dashboard; KPI cards	**M6**	Pilot testing with university drill exercises; UX refinements
M7	FastAPI backend; scenario and optimization API endpoints	**M8**	Government/hydrology API integration prototypes	**M9**	NLP pipeline for Vietnamese/English SOS classification
M10	Performance tuning; solver benchmarking (annealing vs greedy)	**M11**	Second Central Vietnam scenario added to library	**M12**	Year 1 security review; deployment documentation
M13	Live hydrology and traffic data adapters	**M14**	Real-time road closure feed integration	**M15**	Drone imagery ingestion pipeline for flood extent
M16	Provincial partnerships—Thua Thien Hue pilot expansion	**M17**	Multi-province scenario templates (Quang Tri, Quang Nam)	**M18**	Edge deployment option for offline drill nodes
M19	Cloud QPU integration (QAOA / D-Wave trials)	**M20**	Quantum solver A/B testing vs classical baselines	**M21**	Multi-province operational dashboard
M22	Production deployment; SLA monitoring	**M23**	Regional scale-out across Central Vietnam provinces	**M24**	Full product launch; training materials; SOP integration